



Property Tax Reform: Data and Transparency

Chi Hack Night
September 2026

Fritz Kaegi, *Cook County Assessor*
Nicole Jardine, *Chief Data Officer*

Today's goal: Progress in transparency

Today we're here to share some **transparency** tools and to celebrate the wins *for others* that they have empowered.

First, I'll share how past problems with assessments, and how we fixed it by committing to equity and transparency in the model.

Then, we'll share about how another transparency tool, **PTAXSIM**, has helped others conduct policy analyses and provide tax relief.

Finally, we'll share more about the parcel-level data you can download for free.

Our hope is to inspire you to build technology that helps people, and to promote data transparency.

The problem: Over-taxation of more affordable homes.



Chicago Tribune

PROPUBLICA

The New York Times

OPINION



How Lower-Income Americans Get Cheated on Property Taxes

Many homeowners are paying a total of billions of dollars extra because of inequities in assessing property values.

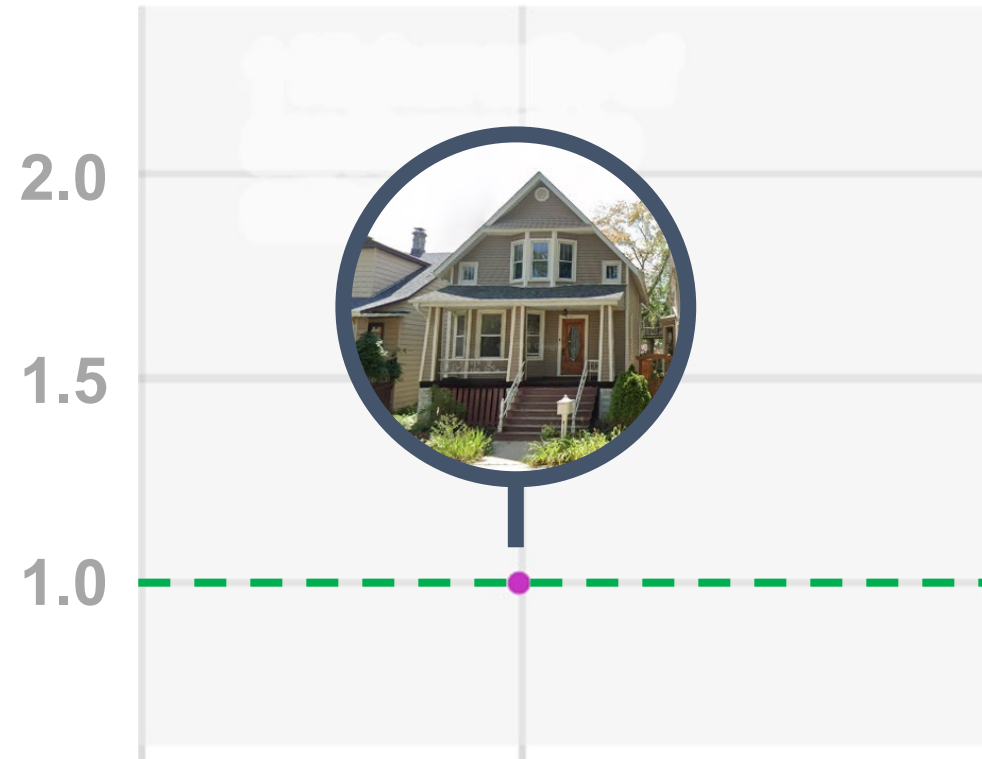
April 3, 2021



2021, based on assessments through 2017.

Chicago

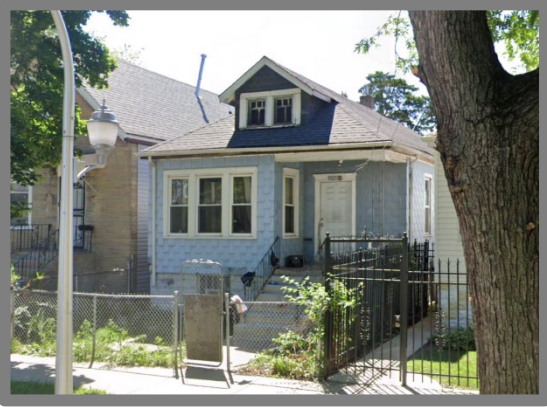
$$\text{Ratio} = \frac{\text{estimated value}}{\text{sale price}}$$



Target ratio for all properties is 1

Source: **Chris Berry/NYT**, using assessments from prior years

Chicago



\$50K sale



\$1M sale

Source: **Chris Berry/NYT**, using assessments from prior years

Chicago



\$50K sale

\$75K estimate

Means a **1.5 ratio**



\$1M sale

\$800K estimate

Means a **0.8 ratio**

Source: **Chris Berry/NYT**, using assessments from prior years

Chicago



\$50K sale

\$75K estimate

Means a **1.5 ratio**



\$1M sale

\$800K estimate

Means a **0.8 ratio**

Source: **Chris Berry/NYT**, using assessments from prior years

Root cause: A model that was inequitable and hidden.

An era of errors

How a bit of computer code fooled taxpayers, flouted legal requirements

BY JASON GROTTO
Chicago Tribune

For more than a decade, the Cook County assessor's office hid a secret inside the massive computer programs used to calculate property tax assessments for single-family homes.

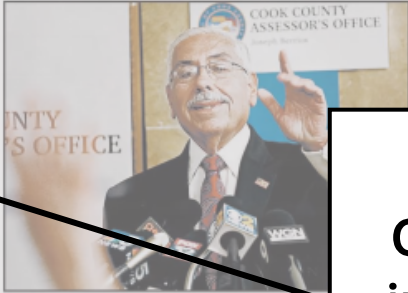
It didn't look like much — just a few snippets of code amid thousands of lines — but it created erroneous valuations for homes throughout the county, affecting the tax bills sent to more than 1 million residential property owners every year.

What the code did was deceptively simple: It decreased every estimated home value in the county by about 40 percent, a troubling practice that ignored legal requirements set out in county ordinances.

The artificially low values threw the property tax system so far out of whack that it may have violated provisions of the state constitution. But, shrouded by an opaque and convoluted assessment process, these widespread inaccuracies

were invisible to the average homeowner.

The Tribune already has revealed how the county's assessment system



NANCY STONE/CHICAGO TRIBUNE

Facing reporters Monday, Assessor Joseph Berrios disputes Tribune findings that property assessments are unfair to the poor. **Page 5**

THE SERIES

Sunday Cook County property taxes harmed the poor, helped the rich

Monday The appeals process makes assessments even less fair

Today An era of errors may have expensive consequences for taxpayers

Read the whole series at chicagotribune.com/taxdivide

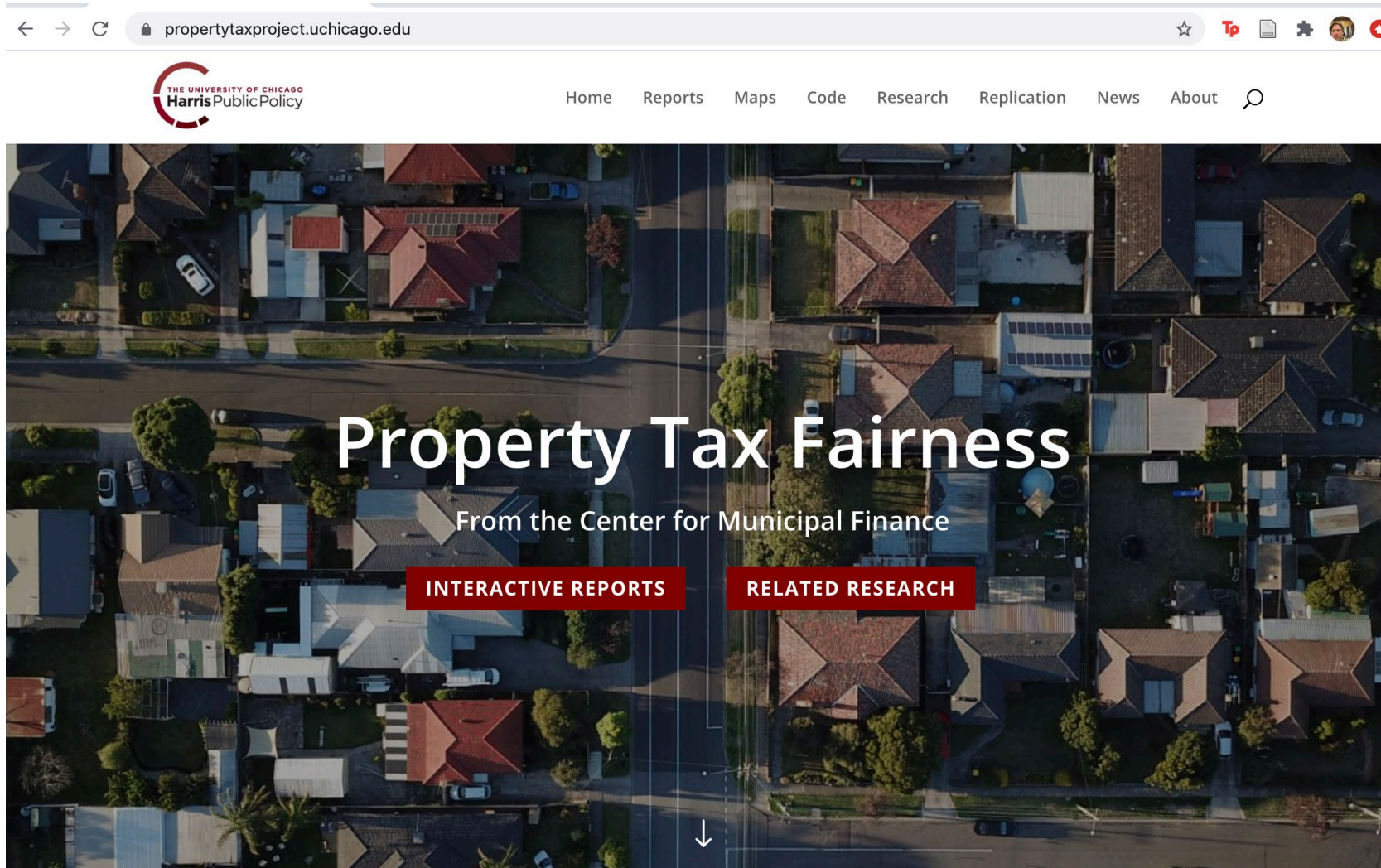
side professional standards while keeping taxpayers in the dark about fundamental problems with assessments. It also exposes the far-reaching consequences when a crucial county agency is allowed to function with little oversight.

Turn to **Property tax**, **Page 6**

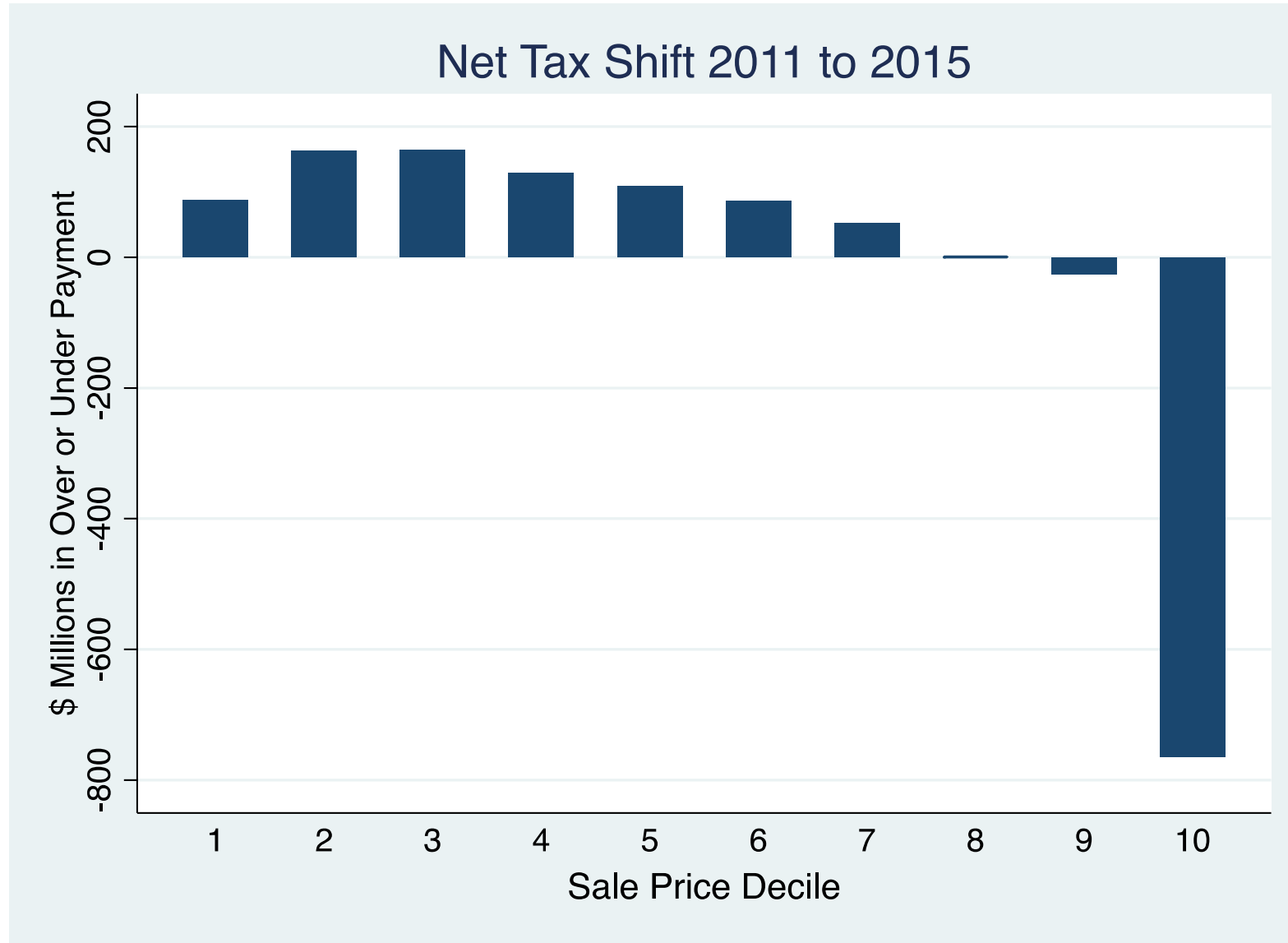
“For more than a decade, the Cook County Assessor’s Office hid a secret inside the massive computer program used to calculate property tax assessments for single-family homes. “

Grotto, Jason. “An era of errors.” Chicago Tribune, June 13, 2017.

The **Center for Municipal Finance** has tracked and quantified regressivity.

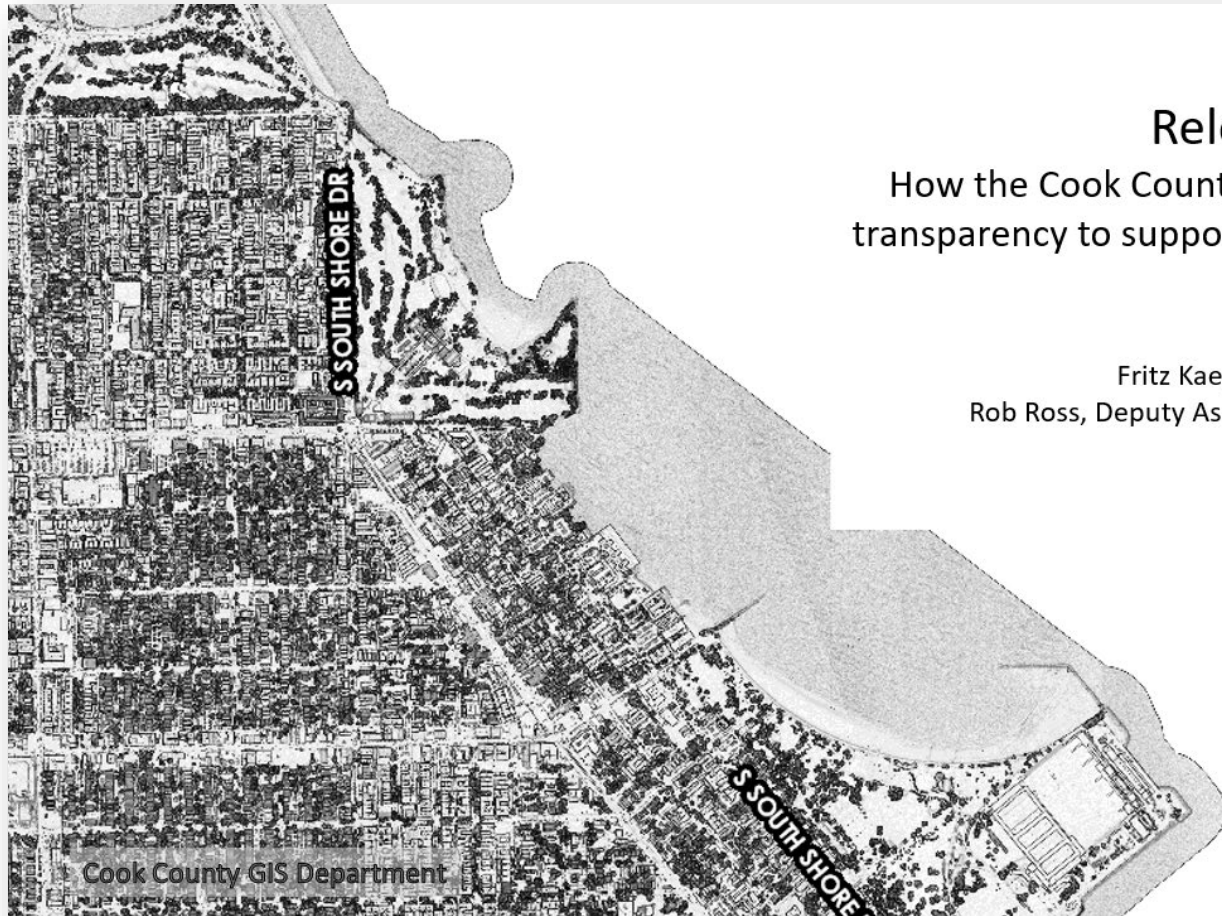


Tax Shifting in Chicago



2019 Commitment: Flatten the curve, and publish the model.

Academics and tech people realized this is a **data** and **modeling** problem – a fixable one.



Release the Code
How the Cook County Assessor is using transparency to support accountability in assessments

Fritz Kaegi, Cook County Assessor
Rob Ross, Deputy Assessor, Chief Data Officer
April 16th, 2019

Cook County GIS Department



~\$2 Billion of Better Property Tax Fairness

An independent, third-party study by the University of Chicago of assessments from 2019 to 2024 found:

*“The Kaegi administration has made substantial progress in improving the fairness of residential assessments. By all measures, **regressivity—the tendency for lower priced properties to be assessed at higher rates, relative to their actual sale price, than higher priced properties—has been dramatically reduced...**”*

*“These improvements have resulted in **over a billion dollars in aggregate tax savings** for owners of middle and lower priced properties...”*

THE UNIVERSITY OF CHICAGO
MANSUETO INSTITUTE
FOR URBAN INNOVATION

THE UNIVERSITY OF CHICAGO
HARRIS SCHOOL
OF PUBLIC POLICY
CENTER FOR MUNICIPAL FINANCE

An Evaluation of Progress on Residential
Assessment Fairness in Cook County

Contact:
Christopher Berry
William J. and Alicia Townsend Friedman Professor
The University of Chicago
crberry@uchicago.edu
This Draft: September 3, 2025

Executive Summary

This report evaluates the changes in property assessments made during Fritz Kaegi's first six years as Cook County Assessor, 2019 to 2024. The central finding is that the Kaegi administration has made substantial progress in improving the fairness of residential assessments. By all measures, regressivity—the tendency for lower priced properties to be assessed at higher rates, relative to their actual sale price, than higher priced properties—has been dramatically reduced, relative to the assessments imposed by Kaegi's predecessor, Joseph Berrios. These improvements have resulted in over a billion dollars in aggregate tax savings for owners of middle and lower priced properties, relative to the over-taxation they experienced during the Berrios years. While areas for improvement remain, Cook County is within industry standards for residential assessment fairness for the first time in years. A second finding is that Kaegi raised average assessments on commercial properties, which would have led to a shift in the aggregate tax burden from residential onto assessments granted by the Cook County Board of Review.

Background

In 2018, Cook County voters elected Fritz Kaegi as assessor. In the months following his election, articles in the *Chicago Tribune* exposing regressivity in predecessor, Joseph Berrios. Regressivity describes a situation where properties are systematically assessed at more than their market value. Kaegi sought to fix regressivity and other deficiencies in the assessor's office, and progress on the fairness of residential assessments made over the years.

Reduced Regressivity in Residential Assessments

The standard method to evaluate residential assessments is a ratio of a property's sale price to the assessed value. In Cook County, this ratio is the ratio of a property's sale price to the assessed value. A property should be assessed at 10% of its market value. For example, a property with a market value of \$500,000 should have an assessed value of \$50,000. If the assessment system is perfect, due to errors in estimation as well as timing of assessments made and when properties sell, it would be unusual for an assessment ratio to be exactly 10%. While assessment errors are

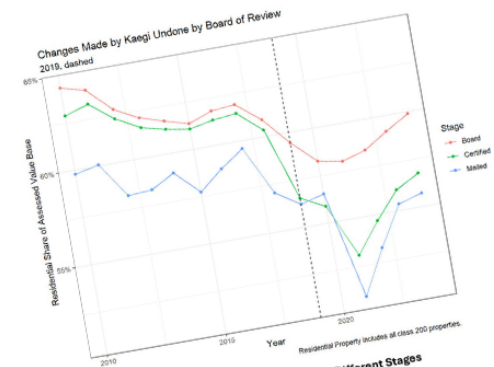
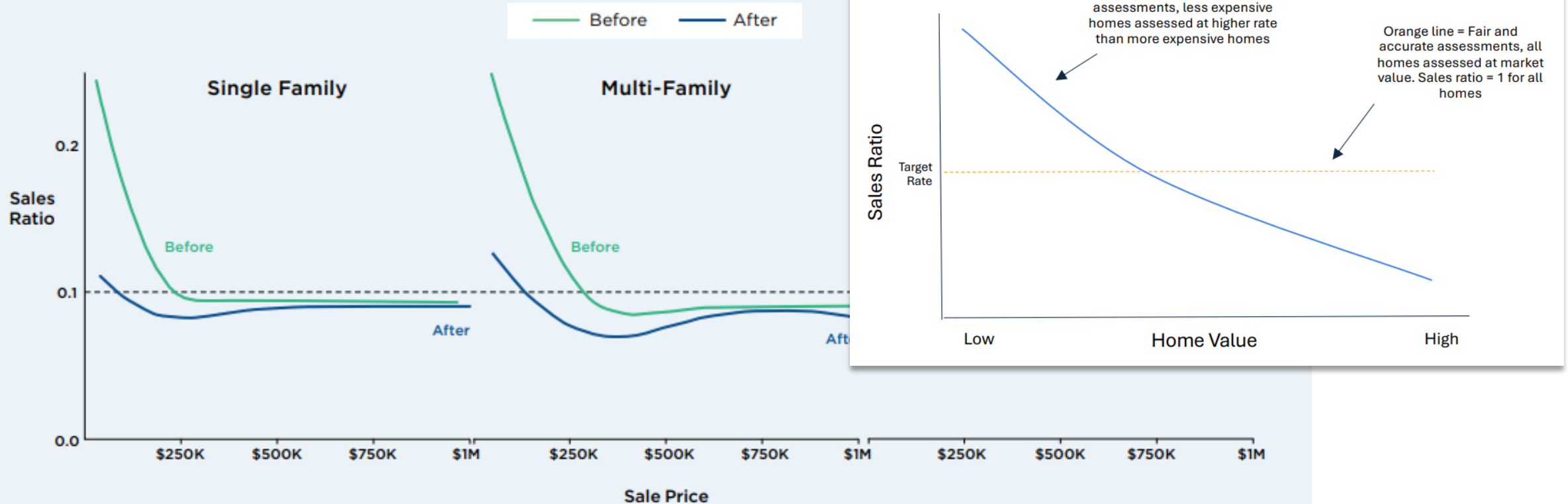


Figure 5A: Residential Share of Assessed Value at Different Stages

~\$2 Billion of Better Property Tax Fairness

EXHIBIT 6 | ASSESSMENT RATIOS BEFORE AND AFTER REFORMS IN COOK COUNTY



**In Cook County, for residential property owners, the assessed value equals 10% of the fair market value. Thus, the ideal assessment ratio in Cook County is 0.10.*

Enduring Commitment to Transparency for Public Service

In 2023, we launched PTAXSIM. What is it, and how has it been used?



[Events](#) [Breakouts](#) [Blog](#) [Projects](#) [About](#) [Membership](#) [Donate](#) [Q](#)

[Events](#) > January 17, 2023

In-person and Online: Why are property taxes in Chicago so high?
(And what can be done to lower them) #524

The thumbnail for the CHI HACKNIGHT event features a map of Chicago on the left, color-coded to show the median percentage change in assessed value (AV) for the 2020-21 tax year. A legend to the right of the map provides the following categories: Decreased (red), 0-10% Increase (light blue), 10-20% Increase (medium blue), 20-30% Increase (dark blue), 30-40% Increase (purple), and Above 40% Increase (dark purple). A red play button icon is centered over the map. On the right side of the thumbnail, there are two video frames: the top one shows a man in a suit speaking into a microphone, and the bottom one shows a man in a dark jacket speaking into a microphone. The CHI HACKNIGHT logo is in the top left corner of the thumbnail. At the bottom left, there is a date and time indicator: "Jan 17, 2023". At the bottom right, there is a "Watch on YouTube" button. The title "Why are property taxes in Chicago so high?" is displayed in large white text at the bottom of the thumbnail.

CHI HACKNIGHT

Why are property taxes in Chicago so high? (And what can be done to lower them)

Median % Change in AV
Tax Year 20-21

- Decreased
- 0-10% Increase
- 10-20% Increase
- 20-30% Increase
- 30-40% Increase
- Above 40% Increase

CHI HACKNIGHT

Jan 17, 2023

Dan Snow, Fritz Kaegi
Cook County Assessor's Office

Why are property taxes in Chicago so high?

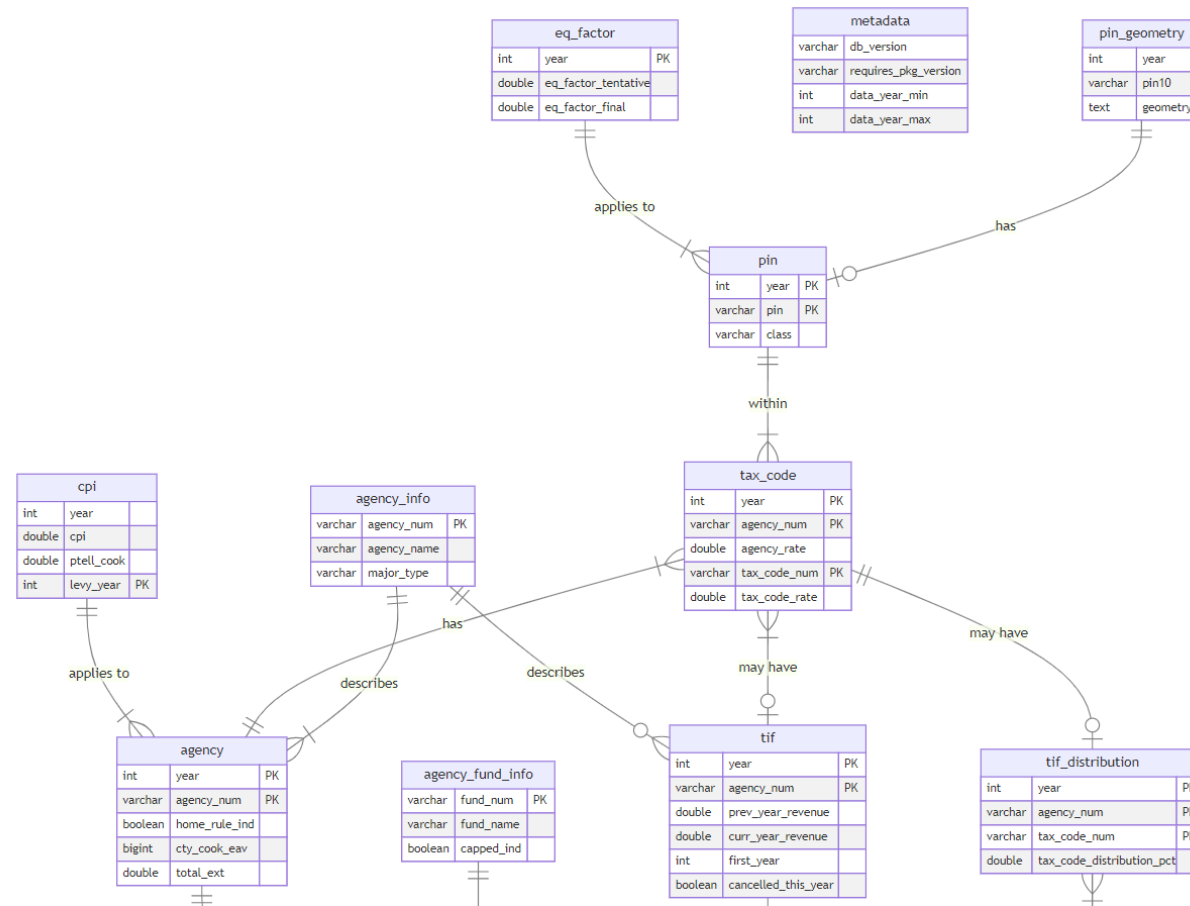
Watch on YouTube

What we built: PTAXSIM



cookcountyassessoril.gov → Data → **Property Tax Simulator Tool**

- **database** of line items for property tax levies and bills, 2006-2024
- **code** for property tax analysis and simulation



#1: UIC and CMAP published policy analyses

<https://propertytaxreform.cookcountyil.gov/>



[How It Works](#) [Reform Group](#) [Our Work](#) [Resources](#) [Newsroom](#)

[Select Language](#)

Building a better property tax system.

The Cook County Property Tax Reform Group strives to build a more predictable, efficient, and equitable property tax system for residents.

Trending Topics

[Explore the property tax process](#)

[Read our study on tax incentives](#)

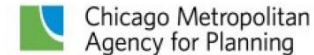
[Property tax reform news](#)

Research Partners



 Chicago Metropolitan Agency for Planning

 Civic Consulting alliance



Current and future use of homestead exemptions in Cook County

Summary

In April 2023, Cook County prioritized three research questions regarding Illinois' homestead exemptions for further study. This memo addresses two of those questions:

- What are the public purposes of Cook County's homestead exemptions? How is success evaluated?
- How does the use of exemptions shift the property tax burden between residential and business properties? Between homeowners and renters? How and why do these shifts vary by taxing district?

The sections below summarize efforts to answer these questions as well as their implications for future policy change. Broadly, the use of exemptions reshapes the local property tax base in communities throughout Illinois such that a portion of homeowners' tax burden is shifted to other taxpayers through higher rates. This design can have greater effect in some communities than others depending on two primary factors: the value of exemptions as a share of the residential tax base (reflecting both the rate of uptake and local home values), and residential property as a share of the total tax base (or more simply, the local land use mix). This analysis shows that areas of Cook County with a more balanced mix of land uses and lower residential property values — including many communities in the South Suburbs — see greater tax burden shifts and higher composite tax rates as a result.

Homestead exemptions are specifically permitted by the Illinois Constitution¹ and primarily enabled by amendments to the Property Tax Code.² But Illinois statutes do not establish what these tools are intended and designed to achieve. A review of past state-level studies and recent legislative proposals suggests that the state lacks a clear and consistent direction for the future of its tax relief programs. If Cook County determines that homestead exemptions have unwanted effects, examples from other states can provide models for reforms. One of the most significant factors shaping exemptions' effects — and the tax savings for homeowners — is which unit of government bears the cost of providing them. Potential changes would need to consider more deliberately what these tools are designed to achieve in Illinois to better align their program design and funding with their public purposes.

¹ Illinois Constitution 1970, Article IX, Section 6.

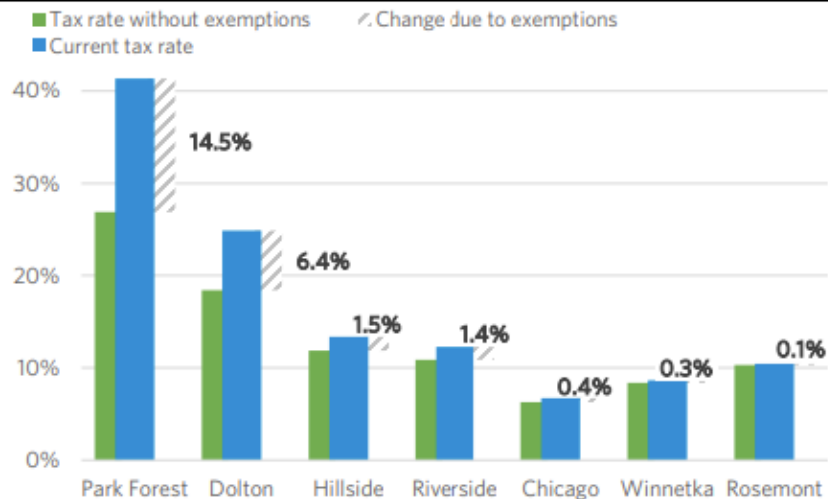
² 35 ILCS 200/15.

#1: UIC and CMAP published policy analyses

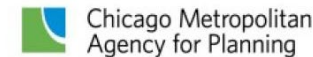
“Research found **unanticipated inequities** in the exemption system.”

Figure 6: Illinois’ homestead exemptions provide tax savings to some property owners by raising the composite tax rates on all properties. The extent of this increase varies within Cook County.

Composite property tax rates with and without homestead exemptions, tax year 2021



Source: UIC analysis of Cook County Assessor’s Office PTAXSIM data.



Current and future use of homestead exemptions in Cook County

Summary

In April 2023, Cook County prioritized three research questions regarding Illinois’ homestead exemptions for further study. This memo addresses two of those questions:

- What are the public purposes of Cook County’s homestead exemptions? How is success evaluated?
- How does the use of exemptions shift the property tax burden between residential and business properties? Between homeowners and renters? How and why do these shifts vary by taxing district?

The sections below summarize efforts to answer these questions as well as their implications for future policy change. Broadly, the use of exemptions reshapes the local property tax base in communities throughout Illinois such that a portion of homeowners’ tax burden is shifted to other taxpayers through higher rates. This design can have greater effect in some communities than others depending on two primary factors: the value of exemptions as a share of the residential tax base (reflecting both the rate of uptake and local home values), and residential property as a share of the total tax base (or more simply, the local land use mix). This analysis shows that areas of Cook County with a more balanced mix of land uses and lower residential property values — including many communities in the South Suburbs — see greater tax burden shifts and higher composite tax rates as a result.

Homestead exemptions are specifically permitted by the Illinois Constitution¹ and primarily enabled by amendments to the Property Tax Code.² But Illinois statutes do not establish what these tools are intended and designed to achieve. A review of past state-level studies and recent legislative proposals suggests that the state lacks a clear and consistent direction for the future of its tax relief programs. If Cook County determines that homestead exemptions have unwanted effects, examples from other states can provide models for reforms. One of the most significant factors shaping exemptions’ effects — and the tax savings for homeowners — is which unit of government bears the cost of providing them. Potential changes would need to consider more deliberately what these tools are designed to achieve in Illinois to better align their program design and funding with their public purposes.

¹ Illinois Constitution 1970, Article IX, Section 6.

² 35 ILCS 200/15.

Research Partners



#2: U Chicago Researchers built a Tax Explainer

<https://miurban.uchicago.edu/resources/>

Property Tax Explainer

[Home](#)

Compare to bill from:

Enter an address or PIN:

Last assessment year ▾

Search

Property Tax Bill Explainer

Key Points

Your tax bill **increased** by **\$243** from 2021 to 2024. An increase in the amount of taxes collected by the government. A decrease in assessed values. A decrease of \$54 was due to other factors.

(C) Change from Commercial-Residential Ratio

Your tax bill may also change when there is a change in how the levy is divided between **all** residential properties compared to **all** commercial properties. When the relative value of commercial property declines, taxes on residential property will go up. In addition, commercial properties are more likely to decrease their assessment value through appeals, which shifts the tax burden more onto residential properties. The share that all residential properties paid in 2024 increased, which caused your tax bill to **increase** by **\$163**.

💡 Change by District

Change in Tax Amount	
Cook County	\$14.39 
Miscellaneous	\$18.43 
Municipality/Township	\$41.45 
School	\$88.82 

What we built: “Tax Spike Dashboard”

cookcountyassessoril.gov → Data → Data Dashboards

Historical Analysis of Property Tax Spikes, 2021-2023

What are the parameters* for a property **tax spike**?
Use the default parameters, or supply your own:

Tax % change*

25%

Tax \$ change*

\$500

Max market value*

\$1,000,000

Parameters: a one-year **tax spike** in 2020-2021, 2021-2022, or 2022-2023 of more than 25% and \$500, with a max market value of \$1,000,000.
Universe of PINs: this dashboard calculates tax spikes among *homeowners*, meaning residential (class 2) PINs that have received at least one homestead exemption, and had the same exemption(s) from 2020-2021, 2021-2022, and/or 2022-2023.

236,841 homeowners

had a **property tax bill spike** between Tax Years 2021-2023.

Total: Cook County

Tax Spike PIN counts, median spike, total spike.

	236,841	\$1,723	\$478.6M
OK	200K	400K	\$1,000
# spike PINs	Median tax spike (\$)	Total tax spike (\$)	

See more: Analyze the data grouped by cities or other districts.

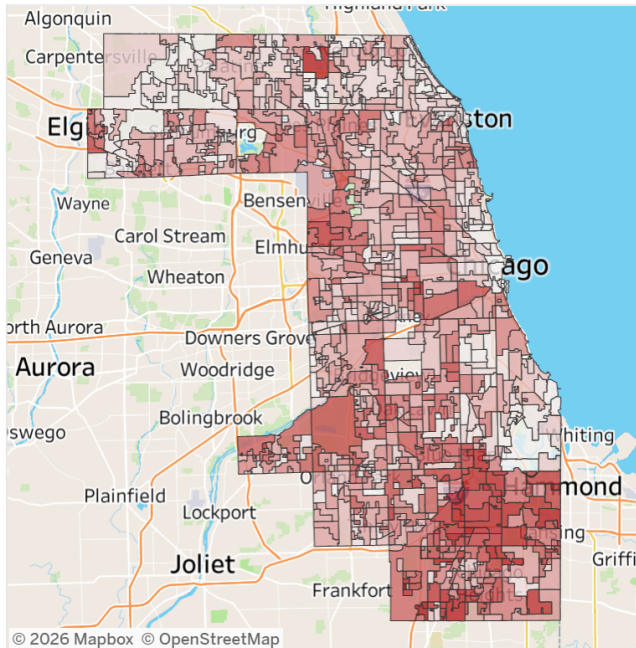
[Tax Spikes per Municipality](#)

[Tax Spikes per Township](#)

[Tax Spikes per State House District](#)

[Tax Spikes per State Senate District](#)

[Tax Spikes per Chicago Ward](#)



What we built: “Tax Spike Dashboard”

Nearly a quarter million homeowners have seen a tax bill spike. Median spike: \$1,723

Historical Analysis of Property Tax Spikes, 2021-2023

What are the parameters* for a property **tax spike**?
Use the default parameters, or supply your own:

Tax % change*

25%

Tax \$ change*

\$500

Max market value*

\$1,000,000

Parameters: a one-year **tax spike** in 2020-2021, 2021-2022, or 2022-2023 of more than 25% and \$500, with a max market value of \$1,000,000.
Universe of PINs: this dashboard calculates tax spikes among *homeowners*, meaning residential (class 2) PINs that have received at least one homestead exemption, and had the same exemption(s) from 2020-2021, 2021-2022, and/or 2022-2023.

236,841 homeowners

had a **property tax bill spike** between Tax Years 2021-2023.

Total: Cook County

Tax Spike PIN counts, median spike, total spike.

	236,841	\$1,723	\$478.6M
OK	200K	400K	\$1,000
# spike PINs	Median tax spike (\$)	Total tax spike (\$)	\$500.0M

See more: Analyze the data grouped by cities or other districts.

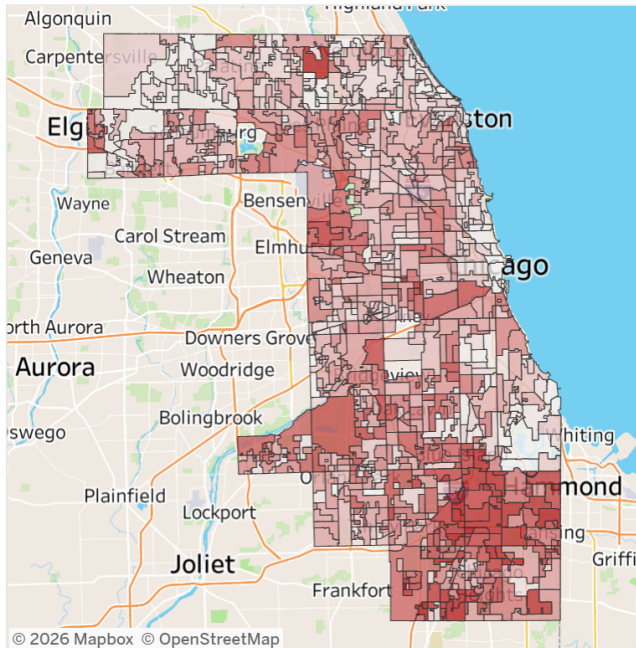
[Tax Spikes per Municipality](#)

[Tax Spikes per Township](#)

[Tax Spikes per State House District](#)

[Tax Spikes per State Senate District](#)

[Tax Spikes per Chicago Ward](#)



What we built: “Tax Spike Dashboard”

We also published the code, documentation, and an export of the data.

Historical Analysis of Property Tax Spikes, 2021–2023

What are the parameters* for a property **tax spike**?
Use the default parameters, or supply your own:

Tax % change*	Tax \$ change*	Max market value*
25%	\$500	\$1,000,000

Parameters: a one-year **tax spike** in 2020–2021, 2021–2022, or 2022–2023 of more than 25% and \$500, with a max market value of \$1,000,000.
Universe of PINs: this dashboard calculates tax spikes among *homeowners*, meaning residential (class 2) PINs that have received at least one homestead exemption, and had the same exemption(s) from 2020–2021, 2021–2022, and/or 2022–2023.

236,841 homeowners
had a **property tax bill spike** between Tax Years 2021–2023.

Total: Cook County
Tax Spike PIN counts, median spike, total spike.

	236,841	\$1,723	\$478.6M
OK	200K	400K	\$1,000
# spike PINs	Median tax spike (\$)	Total tax spike (\$)	

See more: Analyze the data grouped by cities or other districts.

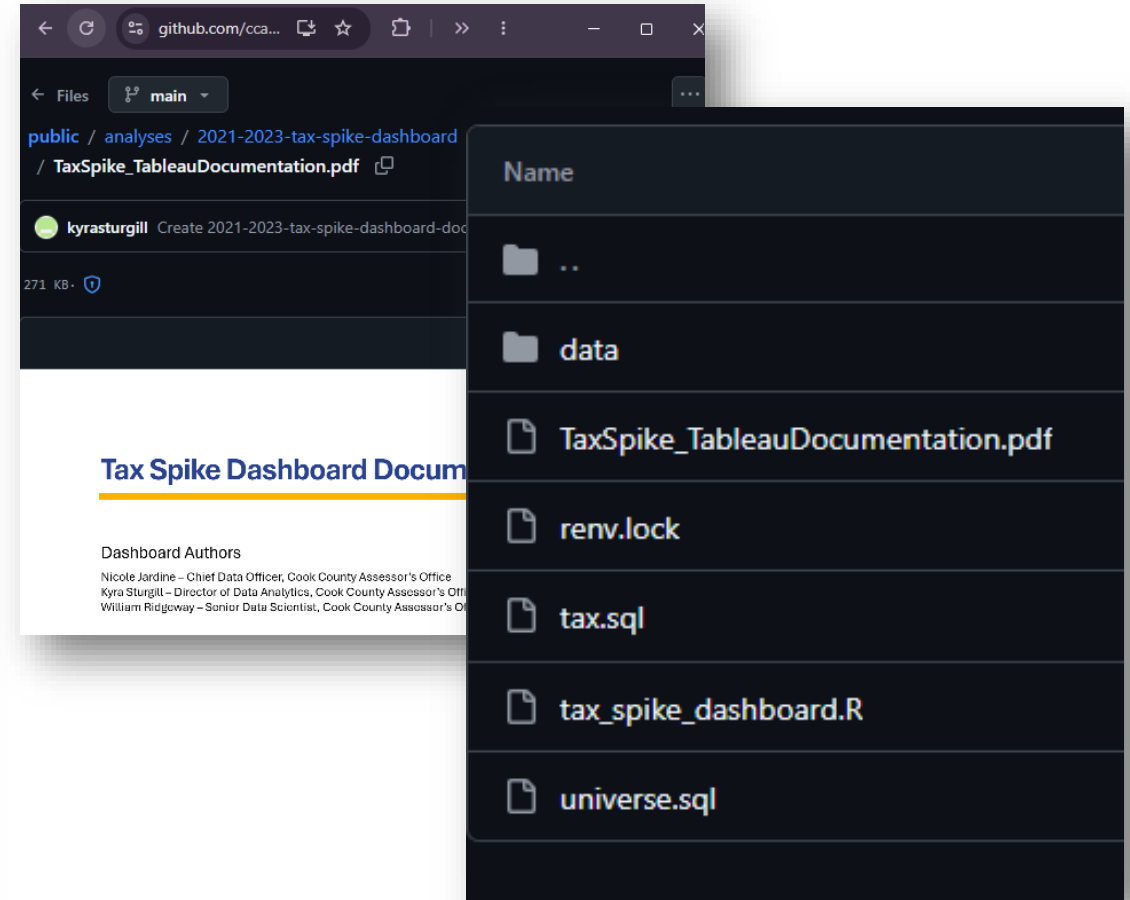
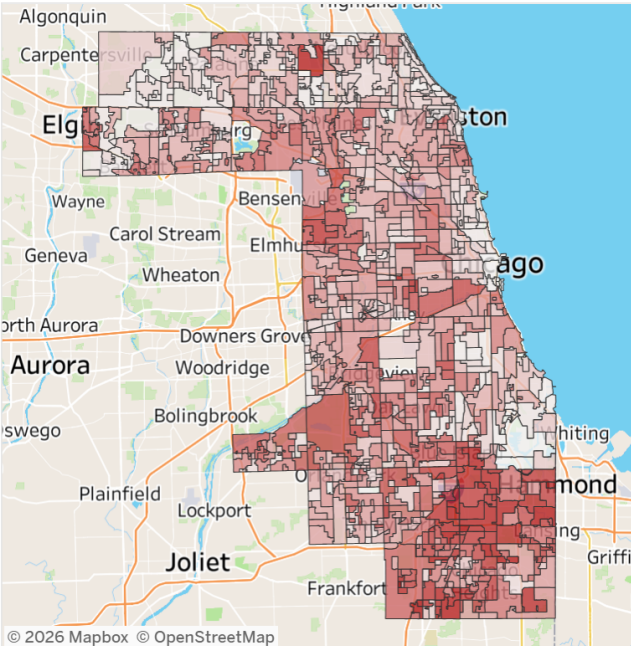
[Tax Spikes per Municipality](#)

[Tax Spikes per Township](#)

[Tax Spikes per State House District](#)

[Tax Spikes per State Senate District](#)

[Tax Spikes per Chicago Ward](#)



#3: Cook County created a Homeowner Relief Fund

Historical Analysis of Property Tax Spikes, 2021-2023

What are the parameters* for a property tax spike?
Use the default parameters, or supply your own:

Tax % change*

25%

Tax \$ change*

\$500

Max market value*

\$1,000,000

Parameters: a one-year **tax spike** in 2020-2021, 2021-2022, or 2022-2023 of more than 25% and \$500, with a max market value of \$1,000,000.
Universe of PINs: this dashboard calculates tax spikes among *homeowners*, meaning residential (class 2) PINs that have received at least one homestead exemption, and had the same exemption(s) from 2020-2021, 2021-2022, and/or 2022-2023.

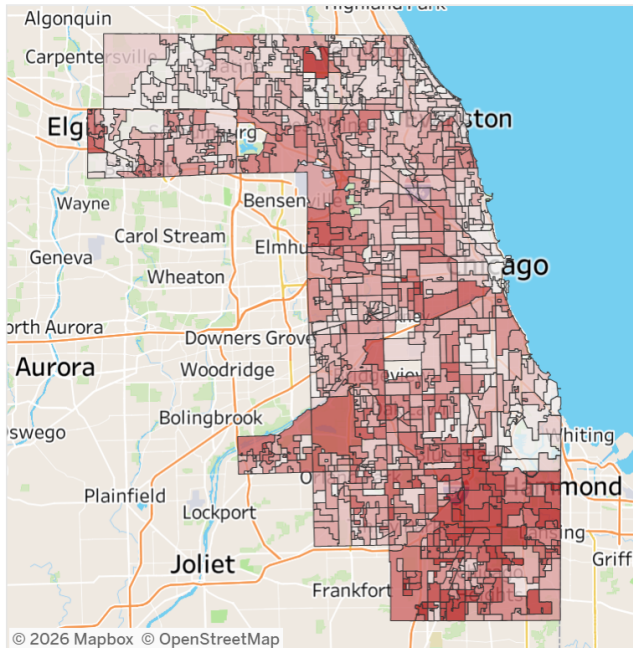
236,841 homeowners

had a **property tax bill spike** between Tax Years 2021-2023.

Total: Cook County

Tax Spike PIN counts, median spike, total spike.

	236,841	\$1,723	\$478.6M
OK	200K	400K	\$1,000
# spike PINs	Median tax spike (\$)	\$500.0M	
		Total tax spike (\$)	



COOK COUNTY
BUREAU OF
ECONOMIC
DEVELOPMENT

Homeowner
Relief Fund

What is the Cook County Homeowner Relief Fund?

The [Cook County Homeowner Relief Fund](#) is a cash assistance program for income-eligible homeowners in Cook County whose property tax bills increased by 30% or more in tax year 2021, 2022 or 2023. The one-time \$1,000 payment is intended to help residents meet basic needs, reduce housing cost burdens and provide financial relief for their households.

This program is funded by the [Cook County Bureau of Economic Development](#) and administered by [AidKit](#).

[Tax Spikes per Chicago Ward](#)

#3: Cook County created a Homeowner Relief Fund

Historical Analysis of Property Tax Spikes, 2021-2023

What are the parameters* for a property tax spike?
Use the default parameters, or supply your own:

Tax % change*

25%

Tax \$ change*

\$500

Max market value*

\$1,000,000

Parameters: a one-year tax spike in 2020-2021, 2021-2022, or 2022-2023 of more than 25% and \$500, with a max market value of \$1,000,000.
Universe of PINs: this dashboard calculates tax spikes among homeowners, meaning residential (class 2) PINs that have received at least one homestead exemption, and had the same exemption(s) from 2020-2021, 2021-2022, and/or 2022-2023.

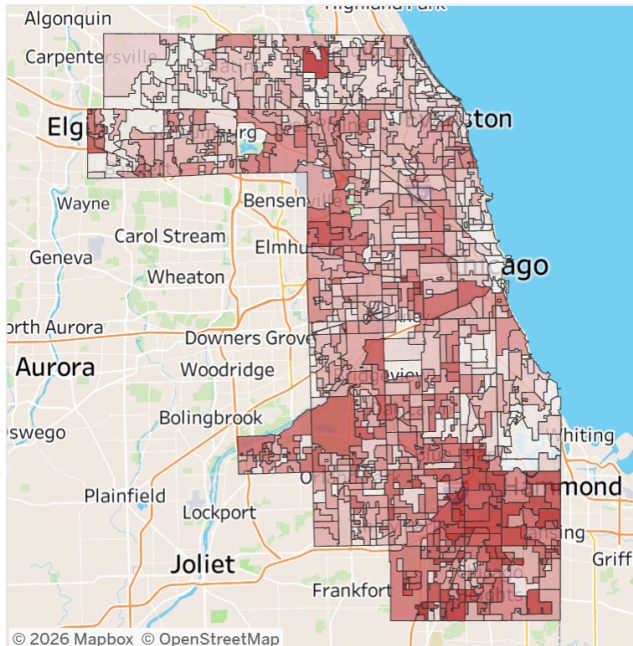
236,841 homeowners

had a **property tax bill spike** between Tax Years 2021-2023.

Total: Cook County

Tax Spike PIN counts, median spike, total spike.

	236,841	\$1,723	\$478.6M
OK	200K	400K	\$1,000
# spike PINs	Median tax spike (\$)	Total tax spike (\$)	\$500.0M



COOK COUNTY
BUREAU OF
ECONOMIC
DEVELOPMENT

Homeowner
Relief Fund

What is the Cook County Homeowner Relief Fund?

The [Cook County Homeowner Relief Fund](#) is a cash assistance program for income-eligible homeowners in Cook County whose property tax bills **increased** by 30% or more in tax year **2021**, **2022** or **2023**. The one-time **\$1,000** payment is intended to help residents meet basic needs, reduce housing cost burdens and provide financial relief for their households.

This program is funded by the [Cook County Bureau of Economic Development](#) and administered by [AidKit](#).

[Tax Spikes per Chicago Ward](#)

PTAXSIM is just one of our transparency commitments



1 Transparency Tool = 3 successes

After publishing PTAXSIM in 2023:

- Academics at multiple university institutions used it for **research**
- A research institute created a **property tax explainer**
- Other government agencies have used it for **tax relief policy**



What other data is out there for you?

Commercial Valuation Data

cookcountyassessoril.gov → Data → Valuation Reports

Methodology Worksheets

Below are the methodology worksheets for each income-producing PIN (Property Assessor's goal in publishing this data is for property owners to understand the ap

There is a Summary tab and a separate methodology tab for each property group.

Methodology Worksheets

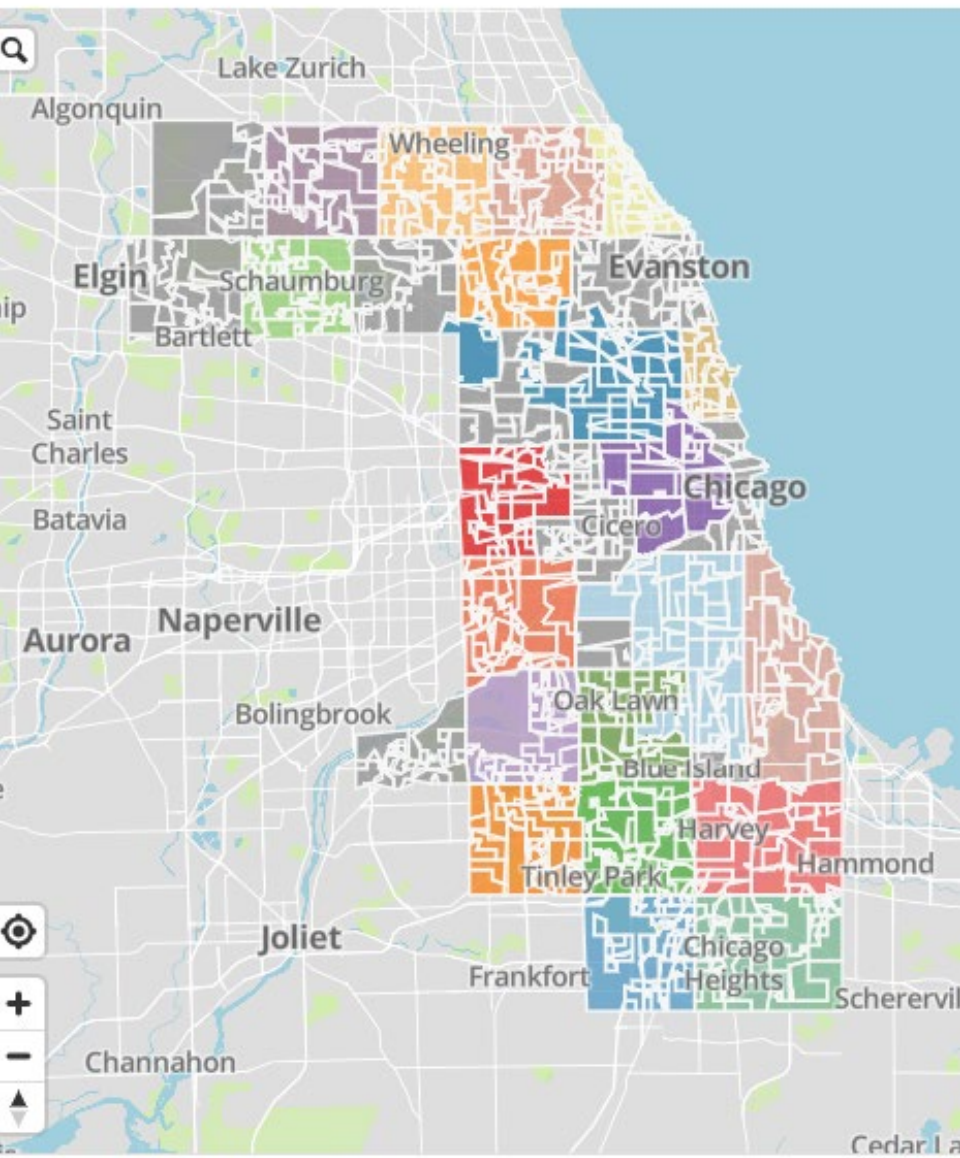
In 2021, we started to publish methodology worksheets for commercial valuations for each township. This level of transparency is unparalleled in large jurisdictions.

But to find the details for a property, it does require you to know a property's PIN and township just to know which file to download.

Wouldn't it be nice to query *all* data?

KeyPIN	Address	Tax Dist	Year Built
16-01-307-001-0000	1057 N FRANCISCO CHICAGO	77001	1996 1996
16-01-307-002-0000			1996 1996
16-01-307-003-0000			1996
16-01-307-004-0000			
16-01-307-005-0000			
14-31-200-029-0000	2211 N ELSTON CHICAGO	77135	1922 1923
14-31-200-030-0000			1912 1912
14-31-200-033-0000			
14-31-200-034-0000			
14-31-501-004-0000			
16-10-411-010-0000	4011 W LAKE CHICAGO	77110	1925
13-25-414-001-0000	2459 W SCHUBERT CHICAGO	77135	1999 1999
13-25-414-002-0000			1999 1999
13-25-414-003-0000			1999 1999
13-25-414-004-0000			1999 1982
13-25-414-009-0000			1999 1999
13-25-414-010-0000			1999 1999
13-25-414-011-0000			1999 1999
13-25-414-012-0000			1999 1999

Assessor - Neighborhood Boundaries



Our Open Data Catalog (starter kit)

cookcountyassessoril.gov → Data → Download Open Data

The screenshot shows the homepage of the Cook County Assessor's Open Data catalog. The header includes the Cook County Government logo, the text "COOK COUNTY GOVERNMENT | OPEN DATA", and a search bar. The navigation menu includes links for Home, Browse All Data, Cook County Website, About Us, User Resources, and Contact Us. The main content area features a large image of the Cook County Courthouse with the title "Cook County Assessor's Open Data". Below the image, there is a paragraph explaining the catalog's purpose and a link to the Cook County Assessor's Office data. A second paragraph describes the data's use in estimating residential property values and creating reports and dashboards. The page also includes social media icons and a "Sign In" button.

COOK COUNTY GOVERNMENT | OPEN DATA

Search

Home Browse All Data Cook County Website About Us User Resources Contact Us

Sign In

Cook County Assessor's Open Data

This is a catalog of [Cook County Assessor's Office](#) data. Below, access data for assessed values, addresses, characteristics, permits, sales, tax-exempt parcel ownership, and other published datasets. Open data promotes transparency and analysis by taxpayers, journalists, researchers, real estate participants, and other government agencies.

This data is published by the Data Department of the Assessor's Office. The Data Department also uses this data to estimate residential property values, conduct sales ratio analyses, and create reports and [dashboards](#) for the public. Documentation for the datasets below and their source code can be found on the [Data Department's wiki](#).

Skip the FOIA line, go straight to the data!

cookcountyassessoril.gov → Data → **Download Open Data**

Single-, Multi-Family & Condo Characteristics

Locations and characteristics of homes, from 1999-present.

Parcel Sales

Sales of all properties, 1999-present.

Assessed Values

Assessed Values, 1999-present.

Commercial Models

Aggregated versions of commercial models.

Parcel Universe, Addresses, Proximity

Location, spatial, address, and proximity data for all parcels in Cook County, 1999-present.

Tax-Exempt Parcels

Locations and owners of tax-exempt parcels.

Permits

Permit data submitted to the CCAO.

Appeals

CCAO appeal filings and decisions.

Example 1: Assessed Values

Assessed Values ★

Assessed values of every parcel for each tax year in Cook County dating back to 1999.

Each row is one:

PIN, per tax year.

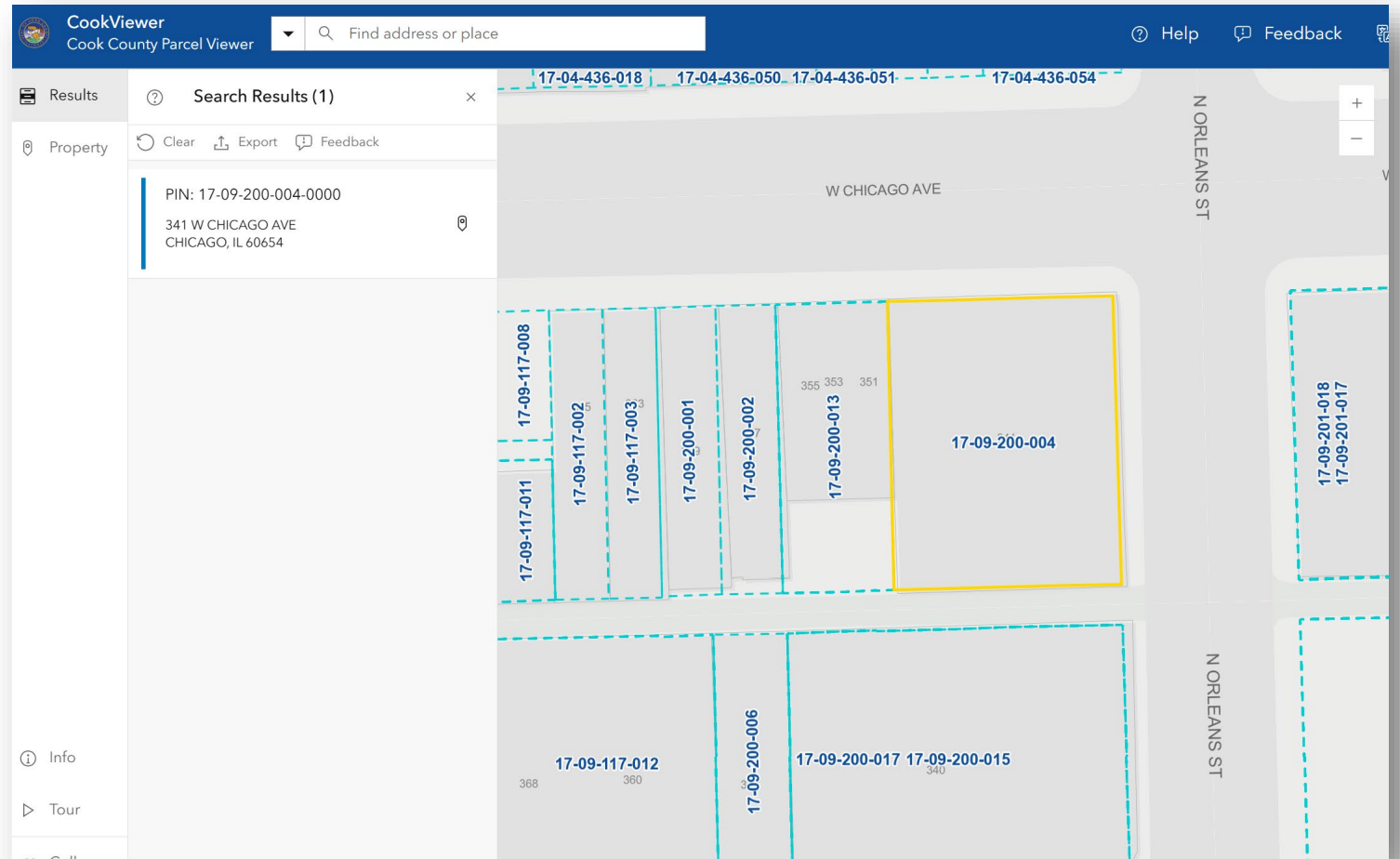
Example use case:

“What were the top 10 highest assessed values in Bloom in 2023?”

Assessor - Assessed Values					
pin	tax_year	class	townsh...	townsh...	neighb...
3221321009	2020	202	12	Bloom	12111
3221403005	2020	EX	12	Bloom	12111
3221404037	2020	EX	12	Bloom	12111
3221411008	2020	EX	12	Bloom	12111
3221418003	2020	EX	12	Bloom	12111
3222104015	2020	890	12	Bloom	12150

Demo: 750 N Orleans

We can use Cook Viewer to search the address and find the PIN



Demo: 750 N Orleans

Then, we can query keypin 17-09-200-004-0000 in Commercial Valuation Data

The screenshot shows the Cook County Government Open Data portal. The header includes the Cook County Government logo and the text "COOK COUNTY GOVERNMENT | OPEN DATA". The navigation bar contains links: Home, Browse All Data, Cook County Website, About Us, User Resources, and Contact Us. Below the navigation bar, there is a "Back to overview" link. The main content area displays a table with the following columns: keypin, pins, year, and township. The data row shows: 17-09-200-004-0000, 17-09-200-004-0000, 2024, and North Chicago. Below the table, there is a "Filters" section with a "Clear all" button. The filters are set to: # year is 2024 and Tr keypin is 17-09-200-004-0000.

keypin	pins	year	township
17-09-200-004-0000	17-09-200-004-0000	2024	North Chicago

Tax Year 2024 data:

Class: 5-91

Adj. rent/sf: \$40

Bldg. sf: 76,800

Cap rate: 0.085

EGI: \$2,352,872

Market Value: \$13,693,000

Example 2: Commercial Valuation Data

Commercial Valuation Data

Each row is one:

KeyPIN, per tax year*.

A KeyPIN is the primary PIN for group of PINs assessed as one economic unit.

Example use case:

“What commercial buildings in Oak Park and River Forest have a bldg. sf over 100,000 sf?”

Assessor - Commercial Valuation Data					
keypin	pins	year	towns...	model...	class(...
01-02-202-C	01-02-202-C	2022	Barrington	5-17s	5-17
01-01-201-C	01-01-201-C	2022	Barrington	5-17s	5-17
01-33-305-C	01-33-305-C	2022	Barrington	5-17s	5-17

Note: a KeyPIN only has a row in its reval tax year. A Barrington KeyPIN only exists in 2022, 2025...

Working with Assessment Data

Tip 1: Beware Excel's Limitations



Excel issues

Excel is limited to ~1M rows.
Cook County has roughly 1.9M parcels.

If you download **all PINs** in a tax year, excel will **not** accurately show or analyze your data.



Excel solutions

Options include:

- Filter your data to reduce rowcount. Class code is a common way to filter for only residential parcels.
- Use google sheets, if allowed. It has a cell cap of 10M. With 1 column of data, you can have 10M rows; and with 2 columns, 5M; so on.

Working with Assessment Data

Tip 2: Know how to join datasets

Joining datasets

Frequently, users have questions that require combining datasets.

Example: “For **Permit PINs**, I want to know their **Assessed Values**.”

General steps

1. Determine the PINs you’re interested in. Download that list.
2. Download additional data that you want to attach to those PINs.
3. Open your first list, and join it to your second list by PIN.

Or: ask us for help!

Cook County Open Data Portal

Tip 3: Pay attention to “Tax Year” versus “Calendar Year”

What’s a Tax Year?

“Tax Year” defines the fiscal tax bill calculation and can be different from the calendar year.

We are in **calendar** year 2026. We are conducting 2026 revaluations, estimating property values in 2026.

This will affect tax bills for Tax Year 2026, which will be issued in **calendar** year 2027.

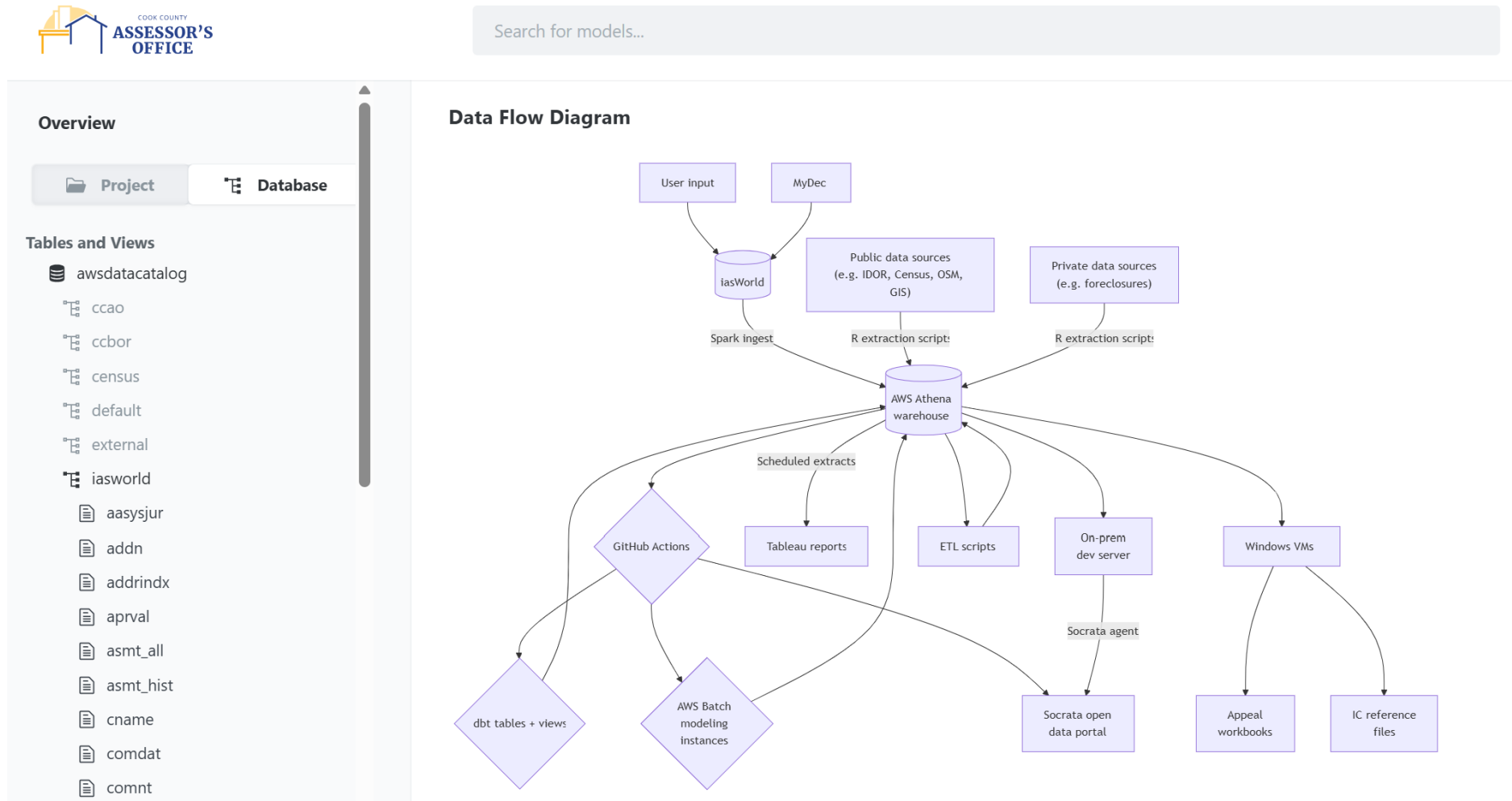
How to use it

Most of our datasets rely on accurate application of “Tax Year”.

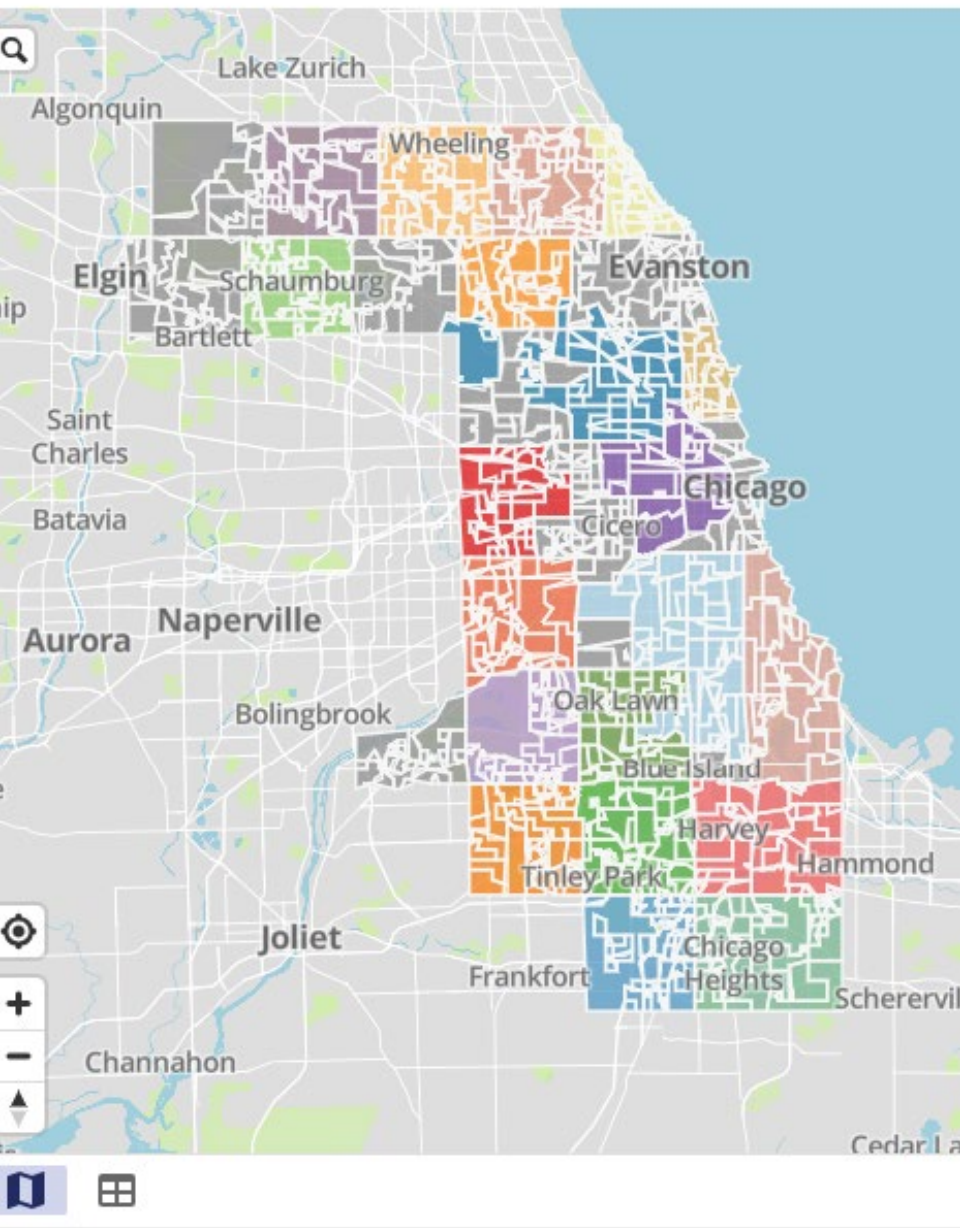
When you join two datasets together, make sure you’re using the same tax year.

Additional transparency tools: Data Catalog

<https://ccao-data.github.io/data-architecture>



Assessor - Neighborhood Boundaries



Our Open Data Catalog (starter kit)

cookcountyassessoril.gov → Data → **Download Open Data**

- Assessed Values ★
- Addresses ★
- Appeals
- Characteristics (Residential)
- Commercial Valuation Data
- Locations ("Universe", "Proximity") ★
- Permits
- Sales
- Tax-Exempt Parcels

★ = For these datasets, every PIN has a row per tax year

External shoutouts: Josh Kalov (GIS) and Jadie Oldfield (BOT)



Why Transparency?

The “Tax Assessor” can be unpopular. Why is transparency worth the effort and risk?

Public have a right to “raw” data *and* consumable reports.

Property assessment is a public function. Parcel-level data should be visible, free, and clearly documented.

Accessible, open data helps the institution.

It reduces FOIA requests and makes data requests easier for us to respond to. It also allows the public, journalists, and researchers to conduct analyses.

Open data powers important policy work.

CCAO Data users have published research papers in academic journals, created **numerous** web apps and services, and identified errors for us to fix.





Fritz Kaegi

Cook County Assessor,
2019–2026 assessments



Questions?



Go forth and hack!

Assessor.Data@cookcountyil.gov
github.com/ccao-data

External shoutouts: Josh Kalov (GIS) and Jadie Oldfield (BOT)